Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your model or result. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

medium: the material through which a mechanical wave travels

01 Warm-up: bring back an earlier idea

A heat engine operates between 500 K and 300 K and absorbs 700 J.
Find maximum ideal work and explain why actual work is usually lower.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

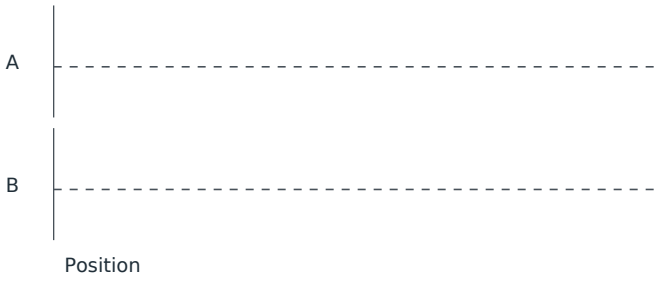
02 Worked example: explain the model

Reference: v = f λ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength 3 m and frequency 2 Hz travels at v=f λ=6 m/s. At the same speed, frequency 3 Hz would require wavelength 2 m.

Your turn: Draw both waves over the same 6 m length and label their frequencies.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

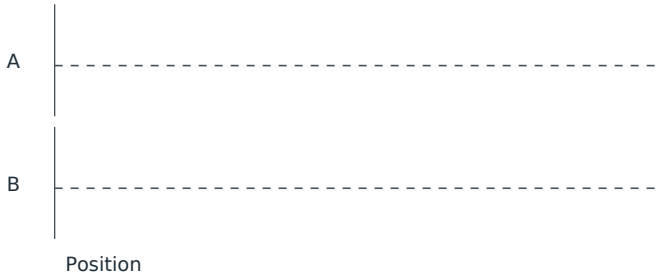
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two waves travel in the same ideal medium at the same speed. One has twice the frequency.
Compare wavelengths.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

04 Apply the physics

A speaker produces a 425 Hz tone in air where sound speed is 340 m/s.
Calculate its wavelength.
Find the frequency that would produce a 0.50 m wavelength in the same air.
Sketch the relative crest spacings.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

A student writes v=f/λ.
Check units and explain why that is not a speed.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

An object at 360 K sits in a vacuum facing surroundings at 240 K. How can it cool, and are absolute-zero surroundings required?
Label temperatures, samples and relevant energy transfers.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Sample A

Sample B

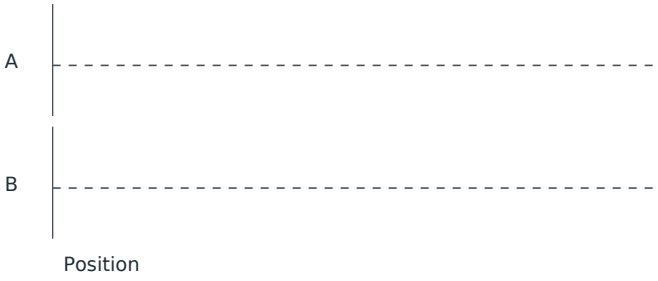
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

A wave travels 12 m in 0.40 s and has frequency 50 Hz.
Find speed and wavelength 0.

Sentence starter: I found ... [unit]. This value means ... in this situation.

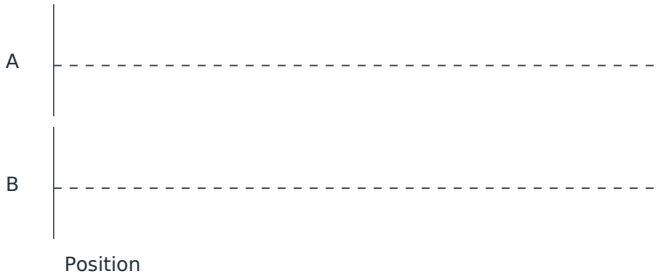


Marzano check - circle: 1 2 3

08 STRETCH: test the model

A wave enters a medium where speed changes from 12 to 8 m/s. Frequency stays 4 Hz.
Find both wavelengths and explain why frequency does not reset at the boundary.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: 1 2 3